

MolSanity: A Reliability Audit for Feature Attributions on Molecular Graph Neural Networks

MolSanity contributors

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<https://github.com/Kar488/molsanity>

Abstract. Feature attributions are routinely used to justify molecular graph neural network (GNN) predictions to chemists, yet they are almost never audited for *reliability*: existing evaluation frameworks ask whether an explanation is faithful to the model, not whether the model’s explanation can be trusted, and not where it stops being trustworthy under the scaffold shift that characterises real drug-discovery workflows. We present **MolSanity**, a reliability-audit framework that wraps canonical attribution implementations (Captum, PyTorch Geometric, RDKit) rather than proposing a new attributor, and scores every (dataset $\times$ backbone $\times$ attributor) cell on six axes: motif-native coherence, occlusion-attribution faithfulness, ground-truth localisation where node labels exist, cross-checkpoint stability, calibration linkage, and confidence/correctness regime stratification. Across 81 committed audit cells spanning 13 datasets, 5 backbones and 6 attributors, the audit shows that *faithfulness is not correctness*: in 11 of the 29 cells where a node-level ground truth exists, an attribution that tracks the model positively (occlusion $\rho > 0$) is nonetheless anti-aligned with the true motif (GT AUROC < 0.5). On MUTAG under a Bemis–Murcko scaffold split, Saliency and Input $\times$ Gradient are not distinguishable from Integrated Gradients in faithfulness ($\rho = 0.376, 0.398$ vs. 0.414 ; paired Wilcoxon $p = 0.206$ and $p = 1.000$) while localising the mutagenic nitro motif at GT AUROC 0.026 and 0.042 against IG’s 0.540 . Ranking attributors by field-standard fidelity recovers the ground-truth-best method in distribution ($\rho(\text{faithfulness, truth}) = 0.943\text{--}0.771$, no mismatch, $n = 120$) but fails under shift, where GraphFrame characterisation selects the worst-localising attributor of six (paired Wilcoxon $p = 0.0001$) and the rank correlation collapses to -0.486 . The attributor ordering itself reverses between regimes ($\rho = -0.714$ across six attributors), and the backbone ordering does not transfer between datasets ($\rho = 0.100$). We therefore argue that an attribution should never be reported without the regime it was audited in. Results reflect the latest committed run: 43/88 cells of the planned grid are complete, the remainder are marked pending and left blank — never imputed. Code, configs, checkpoints and every number in this paper are in the repository.

■ Introduction

A medicinal chemist is shown a graph neural network’s prediction that a candidate is hERG-active, together with a heat map over its atoms. The heat map concentrates on a piperidine nitrogen; the chemist finds that plausible, and the series is deprioritised. Nothing in that workflow ever established that the highlighted atoms are the atoms the model actually used, that they are the atoms that matter chemically, or that the highlighting behaves the same way on the next scaffold. Each of those is a separate question, and the answers can come apart.

They come apart in a specific, measurable way. A saliency map can be *faithful* — the atoms it ranks highest really are the atoms whose removal moves the model’s output most — and simultaneously *wrong* — those atoms are not the substructure that determines the property. On MUTAG under a Bemis–Murcko scaffold split, we measure Saliency at occlusion faithfulness $\rho = 0.376$, not distinguishable from Integrated Gradients at 0.414 on the same molecules (paired Wilcoxon $p = 0.206$, $n = 20$), while its agreement with the mutagenic nitro

motif is GT AUROC 0.026 against IG’s 0.540 — a gap of more than half an AUROC point at equal faithfulness, in the direction that matters (chance is 0.5 ; below 0.5 the attribution is systematically *anti*-aligned with the true motif). A benchmark that scores only faithfulness rates both methods as fine.

This is not an argument that faithfulness metrics are wrong. It is an argument that they are answering a different question from the one a chemist is asking, and that the gap between the two questions widens exactly where molecular property prediction operates: out of distribution, on scaffolds the model has not seen. Molecular property prediction is deployed under scaffold shift almost by construction — the point of a screening model is to score series the training set does not contain — and the standard evaluation protocol for explanations does not stratify by that shift at all.

Contributions. This paper presents **MolSanity**, a reliability-audit framework for feature attributions on molecular GNNs. It is not a new attribution method: it wraps the canonical implementations (Captum [17],

PyTorch Geometric [18], RDKit [19]) and audits them. Specifically:

- We define a six-axis audit over a motif-native (RDKit) decomposition: coherence, occlusion-attribution faithfulness, ground-truth localisation, cross-checkpoint stability, calibration linkage and confidence/correctness regime stratification, with field-standard Fidelity $\pm$, sparsity, the GraphFramEx characterisation score [10] and the PyG/DIG unfaithfulness metric [11] computed on the *same* molecules for direct comparability (The MolSanity audit).
- We run it over 81 committed (dataset $\times$ backbone $\times$ attributor $\times$ split) cells covering 13 datasets, 5 GNN backbones and 6 attributors, and show that faithfulness and correctness dissociate: 11 of the 29 cells that have a node-level ground truth are positively faithful yet anti-aligned with the truth.
- We quantify the consequence for benchmarking practice. Ranking attributors by a faithfulness/fidelity metric — what current evaluation frameworks emit — recovers the ground-truth-best attributor in-distribution, but under scaffold shift the field-standard scores select an attributor the ground truth ranks last (paired Wilcoxon $p = 0.0001$), and the faithfulness-truth rank correlation collapses from +0.771 to -0.486.
- We show the reliability ordering is not a property of the attributor alone: the attributor ranking reverses between the synthetic and the molecular regime ($\rho = -0.714$, $n = 6$ attributors), and the backbone ranking does not transfer between datasets ($\rho = 0.100$, $n = 5$ backbones). No attributor and no backbone is universally reliable.
- We report the negatives with the positives: degenerate cells, cells where the correctness axis simply does not exist, a systematically anti-faithful regression regime, and dependencies that could not be installed. Every number in this paper is generated from committed artifacts by scripts in `paper/figs/`; cells that have not finished are blank, not imputed.

Scope of this draft. The grid is still running. This version reports the 81 cells committed as of the latest run; 43/88 cells of the planned sweep (`configs/full.yaml`) are complete. *Experimental setup* states exactly what is missing, and *Limitations and threats to validity* treats it as a limitation rather than a footnote.

■ Related work

Attribution methods. The attributors audited here are the standard ones. Gradient-family methods — Saliency [2], Input $\times$ Gradient, Guided Backpropagation [3] and Integrated Gradients [1] — are model-agnostic and cheap. Graph-specific methods learn a mask over the input: GNNExplainer [4] optimises a soft edge/feature mask per instance, PGExplainer [5] amortises that into a parametric mask network, and SubgraphX [6] searches connected subgraphs with Monte-Carlo tree search under a Shapley objective. Graph analogues of CAM/Grad-CAM were introduced by Pope et al. [7]; Yuan et al. [8] survey the space. We wrap, never reimplement.

Evaluating explanations. Two evaluation traditions exist. *Faithfulness* measures whether the explanation tracks the model: Fidelity $\pm$ and sparsity [8], the GraphFramEx characterisation score that combines them [10], and the unfaithfulness metric shipped with DIG/PyG [11]. *Correctness* measures whether the explanation recovers a known rationale, which requires ground-truth substructures — typically synthetic [9]. The tension between the two is known: Faber et al. [13] argue that comparing against ground truth is misleading when the trained model does not use that rationale, and Sanchez-Lengeling et al. [14] built the first systematic attribution benchmark for molecular GNNs on exactly this basis. Outside graphs, sanity checks [15] and retraining-based benchmarks [16] made a parallel case that plausible-looking attributions can be uninformative. MolSanity’s position is that neither axis subsumes the other and that the *joint* distribution — and how it moves under scaffold shift — is the object of interest.

Frameworks, and the gap. GraphXAI [9] supplies synthetic generators with exact ground truth and an evaluation library; GraphFramEx [10] systematises fidelity-style evaluation across explainers and tasks; DIG [11] packages methods and metrics; MolFaith [12] evaluates feature attributions on molecular property models under fidelity-based metrics. Table 1 places MolSanity against them. The claim is deliberately narrow: no individual row of that table is new. What is new is the combination — ground-truth localisation *and* faithfulness *and* cross-checkpoint stability *and* calibration linkage, computed over a motif-native decomposition, stratified by scaffold shift, across five backbones, with the field-standard metrics recomputed on the same molecules so the comparison is head-to-head rather than cross-paper.

Table 1: Where the audit sits relative to existing evaluation frameworks. ✓ = core capability, ~ = partial or possible but not central, ✗ = not a focus. The contribution is the combination, not any single row: ground-truth validation and faithfulness both exist elsewhere, but not jointly with cross-checkpoint stability, calibration linkage and scaffold-shift stratification over a motif-native decomposition.

Capability	GraphXAI	GraphFramEx	DIG	MolFaith	MolSanity
Ground-truth explanation benchmarks	✓ (synthetic)	~	✓	✗	✓ (synthetic + motif-proxy)
Faithfulness / fidelity metrics	~	✓	✓	✓	✓ (reproduced for comparability)
Molecular-motif-native audit (RDKit)	✗	✗	~	~	✓ (SSSR + Murcko + BRICS)
Occlusion-attribution agreement	~	✓	~	✓	✓
Cross-checkpoint stability	✗	✗	✗	✗	✓
Calibration linkage	✗	✗	✗	✗	✓
Scaffold-shift regime stratification	✗	~	✗	✗	✓ (the core differentiator)
Multiple GNN backbones (agnostic)	~	✓	✓	~	✓ (GINE/GCN/GAT/MPNN/AttentiveFP)
Paired statistics (Wilcoxon, bootstrap CI)	~	~	~	~	✓
Wraps canonical implementations (no re-impl)	✓	✓	✓	✓	✓ (Captum/PyG/RDKit)

■ The MolSanity audit

Notation. A molecule is a graph $G = (V, E)$ with node features $X \in \mathbb{R}^{|V| \times d}$ and edge features. A model f maps G to a logit vector (classification) or a scalar (regression); t denotes the explained target. An attributor A produces node attributions $a = A(f, G, t) \in \mathbb{R}^{|V|}$. RDKit decomposes G into motifs $\mathcal{M} = \{M_1, \dots, M_K\}$, $M_k \subseteq V$, from SSSR rings, the Bemis–Murcko scaffold [30] and fragment decomposition; the motif score is $s_k = \sum_{v \in M_k} \max(a_v, 0)$. Working at motif rather than atom granularity is what makes the audit chemically legible: a chemist reasons about a nitro group, not about atom 14.

Coherence. Does the attribution concentrate mass sensibly? We report the Gini coefficient of a , the mass in the top 20% of atoms, the largest connected-component fraction among salient atoms, and the motif top-1 share $\max_k s_k / \sum_k s_k$. Coherence is necessary but never sufficient: an attribution can be perfectly concentrated on the wrong motif.

Occlusion faithfulness. For each motif we zero its atoms’ features through the model’s node-mask multiplier and measure the drop in the target logit, $\Delta_k = f_t(G) - f_t(G \setminus M_k)$. Faithfulness is $\rho = \text{Spearman}(s, \Delta)$ over the motifs of a molecule, plus the top-1 agreement indicator $\mathbb{1}[\arg \max_k s_k = \arg \max_k \Delta_k]$. **Higher ρ means more faithful:** the attributor ranks motifs the way occlusion does. $\rho < 0$ means the attributor ranks motifs in the reverse order of their causal effect on the model — worse than an uninformative explanation. For regression, Δ is taken in output space rather than probability space. We also compute Fidelity+ (target-probability drop when the salient atoms are removed; higher is better), Fidelity− (drop when the non-salient

atoms are removed; lower is better), sparsity, the GraphFramEx characterisation score (their weighted harmonic mean [10]) and the PyG/DIG unfaithfulness metric [11], on the same molecules and the same masks.

Ground-truth localisation. Where a node-level ground-truth mask $g \in \{0, 1\}^{|V|}$ exists, correctness is the AUROC/AUPRC of the normalised a against g . Chance is 0.5; below 0.5 the attribution is anti-aligned. When g is degenerate (all-0 or all-1) the metric is undefined and reported as such — never as 0. Only the synthetic sets carry exact g ; MUTAG carries a chemically motivated nitro/aromatic-nitro SMARTS proxy after Debnath et al. [28], and we label it a proxy everywhere. No molecular dataset in the sweep ships per-atom explanation labels, which is precisely why the synthetic anchor is needed.

Cross-checkpoint stability. An attribution that changes when the model is retrained a little is a weak basis for a chemistry decision. We keep an early checkpoint of the same run and report Spearman($s^{\text{early}}, s^{\text{final}}$) per molecule. This axis is absent from the frameworks in Table 1.

Calibration linkage and regime stratification. Predictions are temperature-scaled on validation [31] and we report test expected calibration error (ECE) alongside the attribution metrics, so a cell’s explanation quality can be read next to whether its confidence means anything. Per molecule, the audit assigns a regime from calibrated confidence and correctness — *confident-correct*, *confident-error* (the dangerous one) and *borderline* — and stratifies the reliability metrics by it, together with the correlation between per-molecule calibration error and reliability. The per-molecule records that back this stratification are written to `artifacts/audit/` and are not committed

to the repository, so this draft reports cell-level aggregates only; the regime-stratified analysis is deferred to the completed run ([Limitations and threats to validity](#)).

Aggregation. Cells are compared with paired Wilcoxon signed-rank tests on shared molecules and bootstrap 95% confidence intervals over molecules. p -values are reported unadjusted, and we treat them as descriptive given the reduced per-cell molecule counts.

Engineering. The pipeline is staged and idempotent: each stage writes a `.done` marker keyed to a content hash of its config, every cell is wrapped so a failure is logged and the run continues, and trained checkpoints are committed so the audit reproduces without retraining. Seeds, library versions, git revision and hardware are recorded in a per-run manifest.

■ Experimental setup

Datasets. Tier-1 provides the ground-truth anchor: **SynthMotifs** and its larger twin **SynthMotifsXL**, generated offline (Barabási–Albert base plus house/cycle motifs) with *exact* node ground truth, and **MUTAG** [28, 29] with the nitro proxy. Tier-2 is MoleculeNet [26]: **BBBP** and **BACE** (classification), **ESOL**, **FreeSolv** and **Lipophilicity** (regression). Tier-3 covers chemistry-meets-medicine endpoints: **ClinTox**, **SIDER** and **Tox21** as single-task binary views of the MoleculeNet panels, and **DILI** (drug-induced liver injury) and **hERG** (cardiotoxicity) via the Therapeutics Data Commons [27]. TDC sets are featurised with the same PyG encoding as MoleculeNet so the backbones are drop-in. **BA-2Motifs** and **ShapeGGen** [9] are attempted and gracefully skipped: the former’s download returns HTTP 403 in this environment, the latter’s install depends on the pre-2.5 PyG compiled extensions. Credential-gated resources are out of scope and are never worked around.

Backbones. GINE [20, 21], GCN [22], GAT [23], MPNN [24] and AttentiveFP [25], sharing a pooling head, calibration and splits.

Attributors. Integrated Gradients, Saliency, Guided Backpropagation and Input×Gradient through CapTum [17]; GNNExplainer and PGExplainer through PyTorch Geometric [18]. SubgraphX [6] requires DIG, which installs but cannot import in this environment (no `torch_sparse` wheel at the pinned torch version), so it is left blocked rather than approximated. PGExplainer is classification-only by construction: its mask network trains against class logits, so it is absent from the regression cells.

	scaffold (shift)	random (in-distribution)
SynthMotifs	10/10	1/10
SynthMotifsXL		6/6
MUTAG	10/10	2/10
BBBP	8/10	0/7
BACE	5/5	0/2
ClinTox	4/4	0/2
SIDER	5/5	0/2
Tox21	4/4	0/1
DILI	4/4	0/1
hERG	5/5	0/1
ESOL	7/7	0/4
FreeSolv	5/5	0/1
Lipophilicity	5/5	0/1
BA-2Motifs	0/1	0/1
ShapeGGen	0/1	0/1

81 audited cells committed; 43/88 of the planned grid complete (grey = no cell planned)

Figure 1: Grid coverage as of the latest committed run. Each tile shows completed cells over planned-or-completed cells for that dataset and split. What is outstanding is overwhelmingly the in-distribution (random-split) reference arm: 41 of the 45 pending cell-runs. Grey = no cell planned for that combination.

Splits and budget. The primary split is a deterministic, class-aware Bemis–Murcko scaffold split [30] (whole scaffold groups move together, so there is no leakage, but every fold is guaranteed to contain every class), with a random split as the in-distribution reference. Imbalanced classification uses inverse-frequency class weighting and AUC-based model selection. All committed cells were produced at a *reduced* CPU budget — 30–60 training epochs, 15–25 IG steps, and 20–30 audited molecules per cell (120 for the powered in-distribution arm) — not at the publication budget defined in `configs/full.yaml` (150 epochs, 50 IG steps, 100 molecules per cell). This matters for absolute numbers and for statistical power, and we flag it wherever it could change a conclusion.

Coverage — what is and is not in this draft. The planned grid is 88 cell-runs ($88 = 44 \text{ dataset} \times \text{backbone} \times \text{attributor cells} \times 2 \text{ splits}$). 43 of them are complete. A further 38 audited cells are committed that lie outside the planned list — attributor sweeps on the Tier-2/3 molecular sets and the powered SynthMotifsXL arm — for 81 committed cells in total: 64 classification rows and 17 regression rows. Figure 1 and Table 2 give the breakdown. Three things follow, and we hold to them throughout. First, 41 of the 45 pending cell-runs are random-split (in-distribution) references, so the scaffold-shift arm — the arm the argument rests on — is the more complete one, but in-distribution comparisons within a dataset are mostly not yet available. Sec-

Table 2: What is still running. The planned grid is `configs/full.yaml`; this draft reports the cells committed so far. Pending cells are left blank everywhere in this paper — never imputed. The overwhelming majority of what is outstanding is the in-distribution (random-split) reference arm; the two synthetic sets whose download is blocked in this environment are listed as blocked, not pending.

dataset	cells committed	cells pending
BA-2Motifs [†]	0	2
BACE	5	2
BBBP	8	9
ClinTox	4	2
DILI	4	1
ESOL	7	4
FreeSolv	5	1
Lipophilicity	5	1
MUTAG	12	8
SIDER	5	2
ShapeGGen [†]	0	2
SynthMotifs	11	9
SynthMotifsXL	6	0
Tox21	4	1
hERG	5	1
total	81	45

[†] blocked in this environment (dependency/download unavailable), not merely unfinished.

ond, no committed cell has been run at the `full.yaml` budget; grid coverage is about which cells exist, not about how well-resourced they are. Third, every pending cell appears as “—” or blank in every table and is absent from every figure. Nothing is interpolated, and no aggregate in this paper is computed over an imputed value.

■ Results

Faithfulness is not correctness

Table 3 lists every committed cell in which attribution *correctness* can be measured at all, and Figure 2 plots the joint distribution. Two observations organise everything that follows.

First, the axes are close to orthogonal in the region that matters. Of the 29 cells with both metrics defined, 14 are anti-aligned with the ground truth ($\text{GT AUROC} < 0.5$), and 11 of those are simultaneously positively faithful to the model. That combination — the lower-right quadrant of Figure 2a — is the failure mode a faithfulness-only audit cannot see, because from the model’s point of view nothing is wrong.

Second, the MUTAG scaffold-split block of Table 3

shows the phenomenon at fixed dataset, fixed backbone and fixed split, so the only thing varying is the attributor. Saliency, InputxGradient and Integrated Gradients have occlusion faithfulness 0.376, 0.398 and 0.414 — a spread of 0.038, and neither gradient method is distinguishable from IG on paired per-molecule tests ($p = 0.206$ and $p = 1.000$) — while their agreement with the nitro motif is 0.026, 0.042 and 0.540. Guided Backpropagation behaves like the other two gradient methods (0.341 faithful, 0.112 correct). The two mask-learning methods land elsewhere: GNNExplainer is less faithful (0.279) but far better localised (0.491), and PGExplainer likewise (0.118, 0.401). Reading the faithfulness column alone would rank the gradient family at the top; reading the ground-truth column reverses that. Both columns are computed on the same 20 molecules and the same trained model.

We are careful about what this does and does not license. MUTAG’s mask is a motif proxy, not annotator ground truth, and $n = 20$; on its own this is a suggestive case study. Its role in the argument is to show the phenomenon on a real molecular dataset, with the synthetic sets — where the mask is exact — carrying the load in the next section.

A fidelity-only ranking picks the wrong attributor under shift

If faithfulness and correctness dissociate, then ranking attributors by a faithfulness metric — which is what an evaluation framework does when no ground truth is available — can select the wrong method. We test this directly on cells where all six attributors are audited on the same molecules (Table 4, Figure 3).

In distribution, faithfulness is an adequate proxy. On SynthMotifsXL with exact node ground truth and $n = 120$ molecules on a random split, all three ranking metrics — our occlusion ρ , Fidelity+ and the GraphFramEx characterisation score — select GuidedBP, which is also the ground-truth-best attributor, and the rank correlation between metric and truth across the six attributors is 0.943, 0.829 and 0.771 respectively. There are 0 mismatches. A fidelity-only benchmark is defensible here.

Under scaffold shift on MUTAG, it is not. The ground truth ranks IG best (0.540). Fidelity+ instead ranks GuidedBP first, whose ground-truth localisation is 0.112; the characterisation score ranks Saliency first, at 0.026 — the worst of the six. Both selections differ from the ground-truth-best attributor on paired per-molecule GT AUROC with median gaps 0.422 and 0.524 (paired Wilcoxon $p = 0.0001$ and $p = 0.0001$, $n = 20$ shared molecules). The rank correlation between faithfulness and truth collapses: -0.371 for Fidelity+ and -0.486 for

Table 3: Tier-1 cells: the only cells where attribution *correctness* is measurable. All committed cells that carry a node-level ground truth — exact for the synthetic sets, a chemically motivated nitro-motif proxy for MUTAG. GT AUROC = 0.5 is chance; below 0.5 the attribution is anti-aligned with the true motif. Occlusion ρ is the attribution–occlusion rank agreement (higher = more faithful to the model). Bold marks the best GT AUROC and the best occlusion ρ within each dataset×split block (emphasis only; values are unaltered).

dataset	backbone	attributor	split	n	acc	AUC	GT AUROC	GT AUPRC	occ. ρ	Fid+	Fid−	stab.
MUTAG	GINE	IG	random	20	0.700	0.781	0.075	0.183	0.587	0.211	0.200	0.847
MUTAG	GINE	Saliency	random	20	0.700	0.781	0.022	0.176	0.541	0.209	0.203	0.870
MUTAG	AttentiveFP	IG	scaffold	20	0.650	0.967	0.132	0.150	−0.880	0.002	0.012	0.716
MUTAG	GAT	IG	scaffold	20	0.750	0.890	0.130	0.151	0.268	0.189	0.247	0.929
MUTAG	GCN	IG	scaffold	20	0.750	0.912	0.203	0.222	0.175	0.123	0.228	0.745
MUTAG	GINE	GNNEspl.	scaffold	20	0.800	0.857	0.491	0.347	0.279	0.113	0.278	0.986
MUTAG	GINE	GuidedBP	scaffold	20	0.800	0.857	0.112	0.144	0.341	0.272	0.229	0.903
MUTAG	GINE	Input×Grad	scaffold	20	0.800	0.857	0.042	0.134	0.398	0.251	0.225	0.816
MUTAG	GINE	IG	scaffold	20	0.800	0.857	0.540	0.405	0.414	0.252	0.245	0.728
MUTAG	GINE	PGExpl.	scaffold	20	0.800	0.857	0.401	0.211	0.118	0.026	0.294	0.839
MUTAG	GINE	Saliency	scaffold	20	0.800	0.857	0.026	0.131	0.376	0.267	0.230	0.931
MUTAG	MPNN	IG	scaffold	20	0.750	0.868	0.356	0.274	0.191	0.263	0.312	0.885
SynthMotifs	GINE	IG	random	20	1.000	1.000	0.590	0.489	−0.026	0.102	0.211	0.753
SynthMotifs	AttentiveFP	IG	scaffold	20	0.600	0.510	0.821	0.453	0.643	0.014	0.072	0.659
SynthMotifs	GAT	IG	scaffold	20	0.900	0.990	0.718	0.549	0.527	0.102	0.047	0.568
SynthMotifs	GCN	IG	scaffold	20	0.900	1.000	0.920	0.687	0.096	0.055	0.040	0.448
SynthMotifs	GINE	GNNEspl.	scaffold	20	1.000	1.000	0.498	0.246	−0.017	0.101	0.099	0.934
SynthMotifs	GINE	GuidedBP	scaffold	20	1.000	1.000	0.966	0.845	0.059	0.118	0.086	0.346
SynthMotifs	GINE	Input×Grad	scaffold	20	1.000	1.000	0.964	0.866	0.014	0.114	0.102	0.692
SynthMotifs	GINE	IG	scaffold	20	1.000	1.000	0.742	0.633	0.073	0.109	0.100	0.656
SynthMotifs	GINE	PGExpl.	scaffold	20	1.000	1.000	0.360	0.220	0.106	0.064	0.126	0.701
SynthMotifs	GINE	Saliency	scaffold	20	1.000	1.000	0.983	0.935	0.022	0.123	0.074	0.509
SynthMotifs	MPNN	IG	scaffold	20	0.850	0.910	0.807	0.400	−0.104	0.041	0.030	0.894
SynthMotifsXL	GINE	GNNEspl.	random	120	1.000	1.000	0.501	0.259	0.033	0.364	0.472	0.571
SynthMotifsXL	GINE	GuidedBP	random	120	1.000	1.000	1.000	0.998	0.529	0.495	0.006	0.753
SynthMotifsXL	GINE	Input×Grad	random	120	1.000	1.000	0.989	0.958	0.393	0.493	0.089	0.732
SynthMotifsXL	GINE	IG	random	120	1.000	1.000	0.987	0.945	0.403	0.493	0.021	0.670
SynthMotifsXL	GINE	PGExpl.	random	120	1.000	1.000	0.269	0.165	−0.227	0.050	0.487	0.657
SynthMotifsXL	GINE	Saliency	random	120	1.000	1.000	0.990	0.962	0.424	0.492	0.127	0.632

characterisation, i.e. mildly *negative*.

One detail must not be glossed. MolSanity’s own occlusion ρ does *not* mis-select on this cell — it ranks IG first, which is the ground-truth-best — but its rank correlation with the truth is -0.029 , indistinguishable from zero. It gets the top of the ranking right and the rest of the ranking wrong. The honest conclusion is not that our faithfulness metric is better than the field-standard ones; it is that *no* faithfulness metric in this experiment carries usable information about correctness under shift, and one of the three happened to land on the right winner.

The design of this experiment is also its main weakness, and we state it here rather than in the limitations only: the in-distribution arm and the shift arm are *differ-*

ent datasets (SynthMotifsXL, synthetic and exact, versus MUTAG, molecular and proxy) as well as different splits. Regime and dataset are therefore confounded. The comparison shows that a fidelity-only ranking is adequate in one setting and misleading in another; it does not isolate scaffold shift as the sole cause. Disentangling the two requires the pending random-split arm on the molecular datasets ([Limitations and threats to validity](#)).

The attributor ranking is regime-dependent

Figure 4 lays the two axes out over the audited grid. On SynthMotifs, where the model reaches 100% test accuracy and provably uses the planted motif, the gradient family localises it almost perfectly — Saliency 0.983,

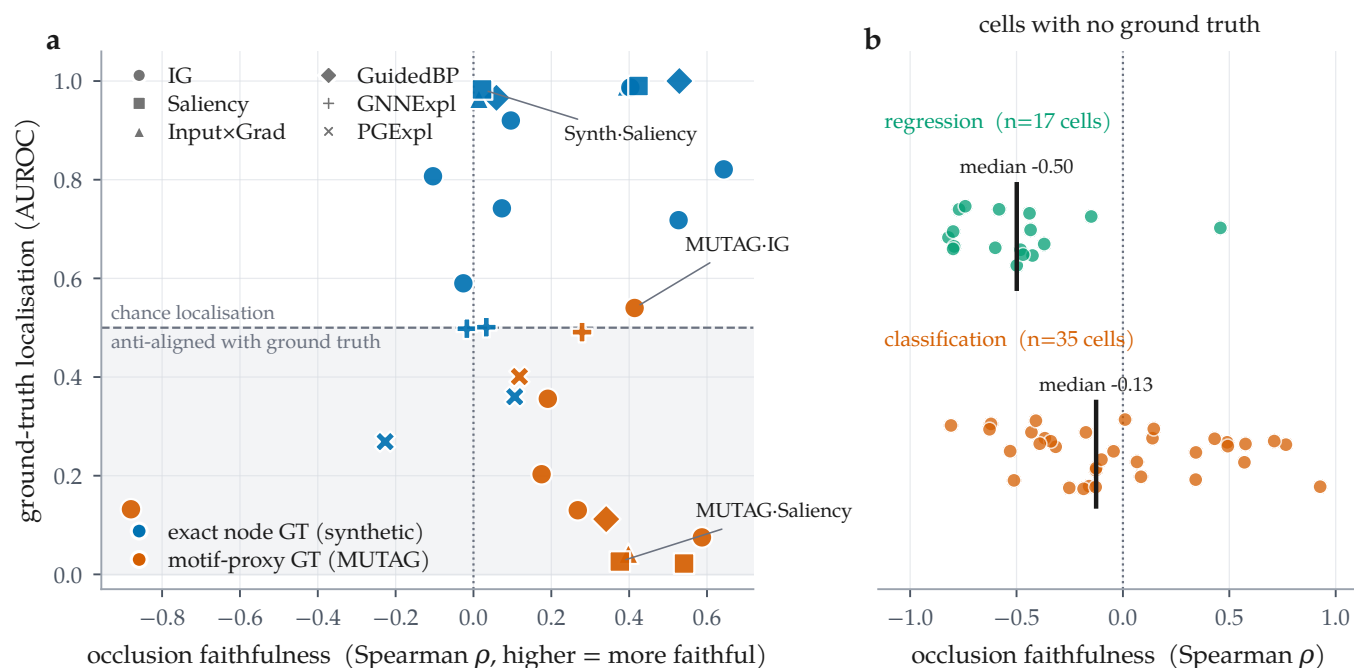


Figure 2: The two axes come apart, and for most cells only one of them exists. (a) Every committed cell that carries a node-level ground truth ($n = 29$), plotted as occlusion faithfulness against ground-truth localisation; colour distinguishes exact synthetic ground truth from the MUTAG motif proxy, marker distinguishes attributor. The shaded band is GT AUROC < 0.5: attributions there are anti-aligned with the true motif. 11 cells sit in the lower-right quadrant — positively faithful to the model and anti-aligned with the truth. (b) The remaining 52 committed cells, on real molecular datasets, have no published per-atom labels: only the horizontal axis is measurable for them. Regression cells (median $\rho = -0.499$) sit almost entirely at negative faithfulness.

Guided Backpropagation 0.966, Input×Gradient 0.964 — while Integrated Gradients trails at 0.742, GNNEExplainer sits at chance (0.498) and PGExplainer below it (0.360). On MUTAG, as above, that ordering is close to inverted. The Spearman correlation between the two GT orderings over the six shared attributors is -0.714 , and between SynthMotifsXL and MUTAG it is -0.600 .

Two mechanisms plausibly contribute, and our data cannot separate them. The synthetic task is one where the model’s rationale and the ground-truth rationale coincide by construction, so any attributor that is faithful is thereby also correct; MUTAG is a task where the model may key on something correlated with, but not identical to, the nitro group — the situation Faber et al. [13] identify as making ground-truth comparison hazardous. That is a reason to report both axes rather than to pick one: an attributor’s apparent quality is inherited partly from whether the model under audit is using the rationale the ground truth encodes.

The practical reading is narrow and defensible: a recommendation of the form “use method X for molecular GNNs” is not supported by any cell in this matrix, because the ranking that supports it is regime-specific.

Backbone dependence

Holding the attributor fixed at Integrated Gradients and varying the backbone (Figure 5) gives a spread of 0.202 AUROC on SynthMotifs (GCN 0.920 down to GAT 0.718) and 0.410 on MUTAG (GINE 0.540 down to GAT 0.130). Both spreads are comparable to the spread across attributors on the same dataset. The orderings do not transfer: $\rho = 0.100$ over the 5 shared backbones. On MUTAG, four of the five backbones put IG’s localisation below chance; only GINE is above it.

AttentiveFP on MUTAG is worth singling out because it fails on both axes at once: occlusion faithfulness -0.880 — strongly *anti*-faithful, the attribution ranks motifs in almost exactly the reverse of their causal order — with GT AUROC 0.132, on a model whose test ROC-AUC is the highest of the five backbones (Table 3). Predictive performance and explanation reliability are not the same axis, and this cell is the clearest demonstration in the matrix.

Cross-task reliability, and the metric that is missing

Table 6 and Tables 7–8 give every committed cell on a real molecular dataset. Three cross-task patterns hold up.

Table 4: Does a faithfulness-only ranking pick the ground-truth-best attributor? Each row ranks the six attributors on the same molecules by one faithfulness/fidelity metric and asks whether its top choice is the attributor the ground truth ranks best. In distribution the answer is yes for every metric; under scaffold shift the two field-standard metrics choose an attributor that is anti-aligned with the ground truth, and the rank correlation between faithfulness and correctness collapses. p is a paired Wilcoxon test on per-molecule GT AUROC between the selected and the ground-truth-best attributor.

regime	cell	ranking metric	its top pick	pick GT	GT-best	GT-best	mismatch	Wilcoxon p	$\rho(\text{faith}, \text{GT})$
in-distribution	SynthMotifsXL · GINE, $n=120$	occlusion ρ	GuidedBP	1.000	GuidedBP	1.000	no	—	0.943
		Fidelity+	GuidedBP	1.000	GuidedBP	1.000	no	—	0.829
		characterisation	GuidedBP	1.000	GuidedBP	1.000	no	—	0.771
scaffold shift	MUTAG · GINE, $n=20$	occlusion ρ	IG	0.540	IG	0.540	no	—	-0.029
		Fidelity+	GuidedBP	0.112	IG	0.540	yes	0.0001	-0.371
		characterisation	Saliency	0.026	IG	0.540	yes	0.0001	-0.486

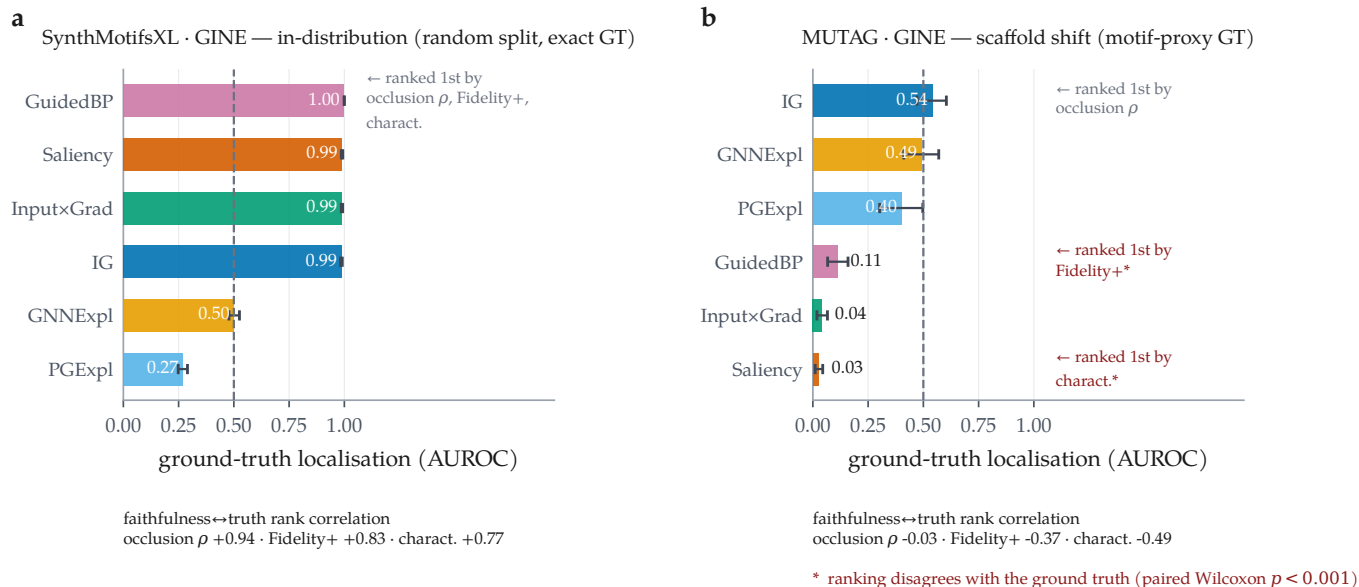


Figure 3: In distribution, faithfulness predicts correctness; under scaffold shift it does not. Ground-truth localisation per attributor, on shared molecules, with bootstrap 95% confidence intervals; annotations mark which attributor each faithfulness-style ranking would have selected. (a) SynthMotifsXL (exact node ground truth, random split, $n = 120$): all three rankings pick the ground-truth-best attributor. (b) MUTAG (motif-proxy ground truth, scaffold split, $n = 20$): Fidelity+ picks GuidedBP (GT AUROC 0.112) and the GraphFrameEx characterisation score picks Saliency (GT AUROC 0.026), against a ground-truth-best of IG at 0.540. Rank correlations between each faithfulness metric and ground truth are given below each panel.

Regression is systematically anti-faithful. Across 17 committed regression cells, 16 have negative occlusion faithfulness, with median -0.499 ; the sole exception is Lipophilicity under GAT. The same cells have well-behaved predictive performance (R^2 from 0.665 to 0.694 on ESOL across 3 backbones). Two readings are available and we cannot yet separate them: either attributions on these regression models genuinely invert the causal ordering of motifs, or the occlusion operator — zeroing node features and reading the shift in a scalar output — is not a well-posed counterfactual in output space, where masking atoms moves predictions toward a pooled mean regardless of which atoms are masked. The negative Fidelity $\pm$ values in Table 6 are consistent with the second reading. We therefore report the regression numbers but do not build the headline on them, and

we flag the probability-space definition of Fidelity $\pm$ as not transferring cleanly to regression.

Coherence is high nearly everywhere, and says little.

Motif top-1 share lies between 0.422 and 0.887 across the 52 molecular cells of Tables 6–8, with 50 of them inside a narrow 0.6–0.9 band, largely independent of whether the same cell is faithful or anti-faithful. Concentration on a single motif is easy to achieve and, on this evidence, close to uninformative about reliability. It belongs in the audit as a descriptive statistic, not as a quality score.

Stability is high but not uniform.

Median cross-checkpoint stability over 81 cells is 0.839, with a minimum of 0.346. By attributor, GNNExplainer is the most stable (median 0.949, $n = 13$) and PGExplainer

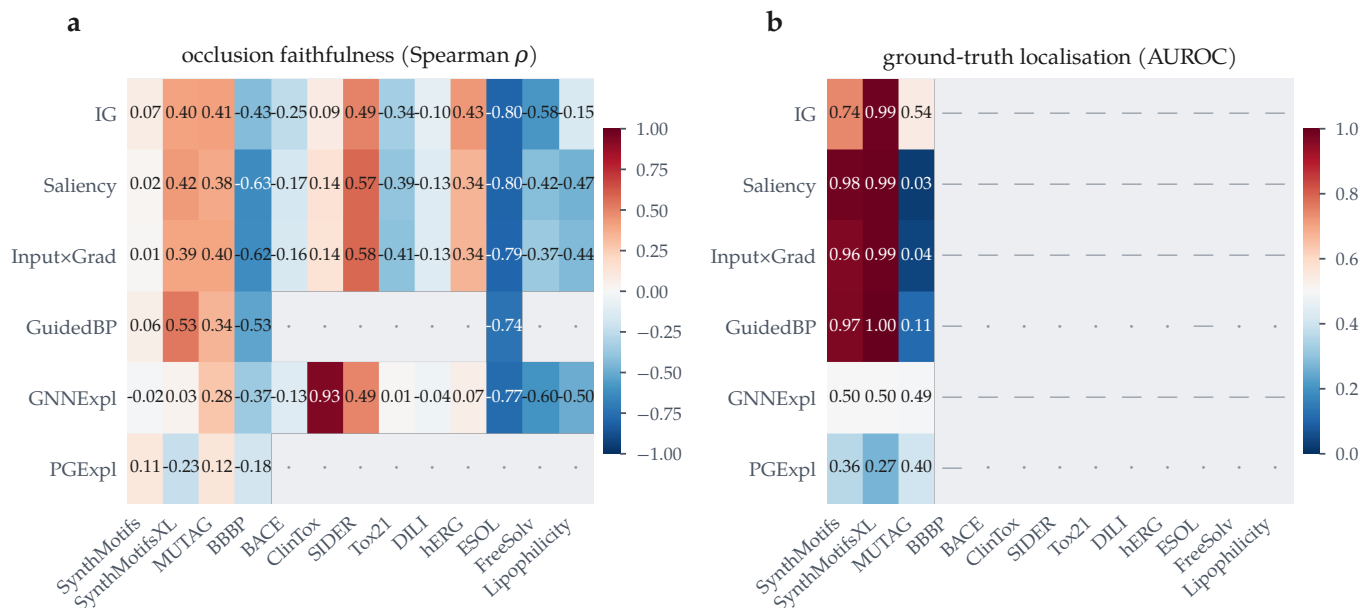


Figure 4: Reliability is a property of the (attributor, dataset) pair, not of the attributor. All committed GINE cells at each dataset’s primary split. (a) Occlusion faithfulness; blue = anti-faithful. (b) Ground-truth localisation; the grey block is not a low score but an *unmeasurable* one — no molecular dataset here publishes per-atom labels. “—” marks an audited cell whose metric is undefined, “.” a cell that has not been audited. The attributor ordering in the three ground-truth columns of (b) is not preserved between them ($\rho = -0.714$ between SynthMotifs and MUTAG).

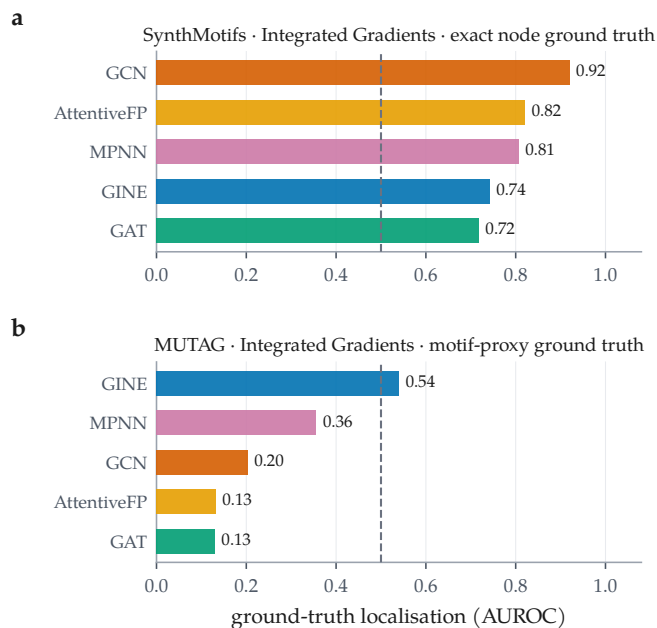


Figure 5: The backbone changes attribution reliability as much as the attributor does. Integrated Gradients held fixed; ground-truth localisation across the five backbones, on the exact-ground-truth synthetic set (a) and the motif-proxy molecular set (b). Dashed line is chance. The two orderings are essentially unrelated ($\rho = 0.100$, $n = 5$).

the least (0.679, $n = 4$), with Integrated Gradients at 0.858 ($n = 32$); the per-attributor counts are unequal, so these medians are descriptive rather than a controlled comparison.

Table 5 shows why per-cell molecule counts matter. On the powered in-distribution cell ($n = 120$) the paired tests separate the attributor families decisively; on the shift cell ($n = 20$) most contrasts are indistinguishable. A framework that ranks attributors on faithfulness at these sample sizes is often ranking noise — an independent reason not to treat a fidelity leaderboard as a method recommendation.

Negative and degenerate results

We record the cells that did not work, because they carry information.

A broken model can be faithfully explained. The lowest-AUC cell in the matrix is BACE $\times$ GCN, with test accuracy 0.133 and ROC-AUC 0.224 — substantially *worse* than chance, i.e. an anti-predictive model. Its attributions are among the most faithful in the entire classification block (occlusion $\rho = 0.766$, characterisation 0.370). Faithfulness is a model-relative property: it certifies that the explanation describes the model, and says nothing about whether the model is worth describing. Any reliability report that omits the predictive metrics can be gamed by exactly this cell.

Majority-class collapse. Four rows in Table 8 report test accuracy ≥ 0.99 with ROC-AUC below 0.9 — all Tox21, a single-task view in which positives are a small minority. Accuracy is the wrong headline there and we re-

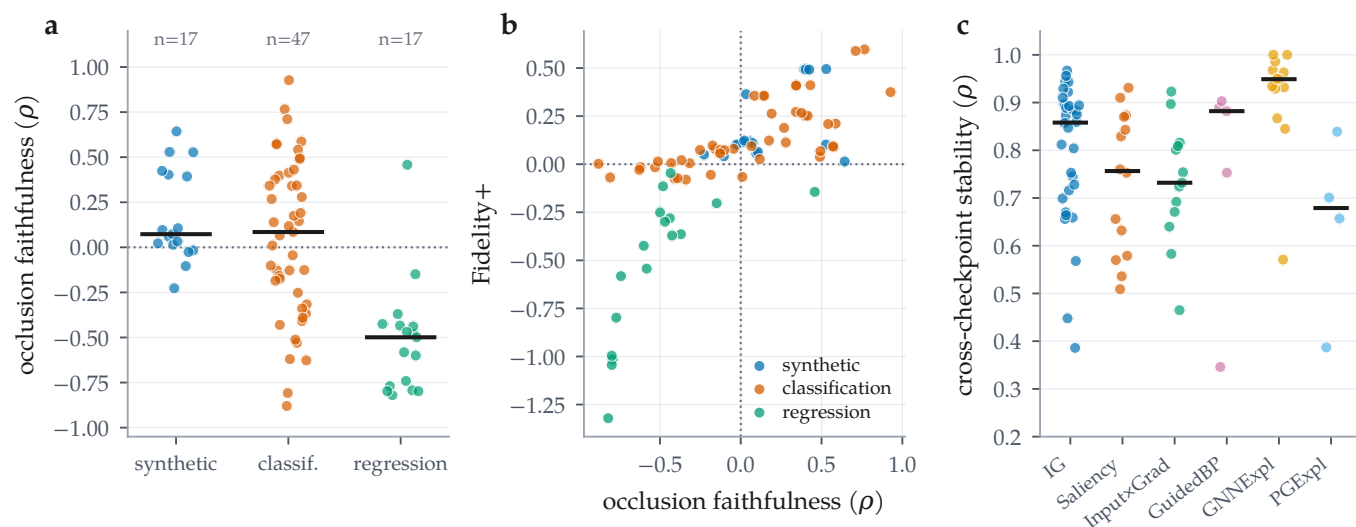


Figure 6: Distributions across all 81 committed cells. (a) Occlusion faithfulness by regime; horizontal bars are medians. Regression cells are almost uniformly anti-faithful (16/17 negative). (b) The field-standard Fidelity+ against MolSanity’s occlusion ρ on the same molecules: related but far from interchangeable, and the regression cells occupy a quadrant of their own. (c) Cross-checkpoint stability by attributor; the mask-learning methods are not uniformly more stable than the gradient ones.

port AUC alongside it throughout. Under the reduced budget, scaffold splits on imbalanced sets could produce near-single-class test folds; the class-aware split and inverse-frequency weighting used for the Tier-2/3 cells are what keep these cells interpretable, and the earlier unweighted pass is retained in the repository rather than deleted.

Calibration varies more than accuracy. Test ECE across the classification matrix has median 0.104 and reaches 0.398. Since the audit’s regime stratification keys off calibrated confidence, cells at the top of that range cannot support a confident-error analysis at all — a prerequisite the audit checks rather than assumes.

Blocked, not silently dropped. SubgraphX (via DIG) and ShapeGGen (via GraphXAI) could not be installed: both need the pre-2.5 PyG `torch_sparse/torch_scatter` compiled extensions, for which no wheel exists at the pinned torch version. BA-2Motifs’ download returns HTTP 403 here. All three are logged as blocked, occupy no row in any table, and are excluded from every aggregate. ShapeGGen’s exact-ground-truth role is covered by the offline SynthMotifs generator; SubgraphX’s absence is a genuine gap in attributor coverage.

■ Discussion

What the audit buys. The single most useful thing MolSanity produces is not a score but a *negative certificate*: for 52 of the 81 committed cells — every real molecular dataset in the sweep — attribution correctness cannot be

measured at all, because no per-atom explanation labels exist. Those cells can be audited for faithfulness, coherence, stability and calibration, and this paper’s central result is that none of those, individually or together, is a reliable stand-in for correctness under shift. Making that gap explicit is more useful than filling it with a metric that looks like correctness and is not.

How attributions should be reported. Three practices follow directly from the matrix. (i) Report the regime. An attribution audited in-distribution carries no warrant out of distribution, and the same attributor moves from best to worst across regimes ($\rho = -0.714$). (ii) Report the backbone. The backbone ordering does not transfer between datasets ($\rho = 0.100$), so “IG on a GNN” is under-specified. (iii) Report the predictive metrics next to the explanation metrics, because a faithful explanation of an anti-predictive model is exactly what the BACE $\times$ GCN cell shows you can get.

Why reliability is regime-, backbone- and dataset-dependent. A faithfulness metric asks about the model; a ground-truth metric asks about the chemistry; they agree only when the model has learned the chemistry. That coincidence is engineered into synthetic benchmarks and is not guaranteed on molecular data, particularly out of distribution. Attribution evaluation inherits the model’s inductive limitations, which is why the correct object of study is the cell — (dataset, backbone, attributor, split) — and not the attributor.

■ Limitations and threats to validity

Incomplete grid. 43 of 88 planned cell-runs are complete. 41 of the 45 pending runs are the random-split in-distribution arm, which is exactly what is needed to turn the fidelity-only ranking experiment into a within-dataset contrast. Until those land, regime and dataset remain confounded in the headline experiment, and the shift claim rests on a synthetic-versus-molecular comparison plus the within-MUTAG faithful-but-wrong result.

Reduced budget. No committed cell was run at the publication budget. Per-cell molecule counts are 20–30 (120 for the powered arm), so per-cell confidence intervals are wide and most paired contrasts on the shift cells are underpowered (Table 5). Absolute values should be expected to move when the full run completes; what we would expect to survive is the qualitative dissociation, and that expectation is itself a claim this draft cannot yet verify.

Proxy ground truth. MUTAG’s mask is a nitro/aromatic-nitro SMARTS proxy, not annotator labels. It is chemically motivated [28] but it encodes an assumption about which substructure ought to matter, and Faber et al. [13] give the general reason that assumption can be wrong: the trained model may use a different, equally valid rationale. Where the distinction changes the reading, exact ground truth comes from the synthetic sets.

Synthetic ground truth is not molecular. SynthMotifs and SynthMotifsXL have exact node labels but are structural, non-molecular graphs. They validate the audit machinery and support the in-distribution arm; they do not establish chemical generalisation.

Single split, single seed. Each cell is one deterministic split at one seed. Cross-checkpoint stability partially probes run-to-run variability, but it is not a substitute for repeated splits, and no confidence interval in this paper covers split variance.

Occlusion as a counterfactual. Zeroing node features takes the graph off the data manifold; a motif whose removal barely moves the output may be redundant rather than unimportant. This is a known weakness of occlusion/fidelity-style metrics generally, and it bears most heavily on the regression cells, where we decline to interpret the sign.

Coverage of attributors. SubgraphX is absent for dependency reasons, and PGExplainer is absent from regression by construction. Conclusions about the pertur-

bation family rest on GNNExplainer and PGExplainer alone.

Multiplicity. p -values are unadjusted across a large number of paired comparisons; individual contrasts in Table 5 should be read as descriptive.

■ Conclusion

Molecular GNN attributions are audited today for whether they describe the model. Across 81 committed audit cells we find that this is not the same question as whether they describe the chemistry, and that the two questions diverge in the regime molecular property prediction actually operates in. In 11 of 29 ground-truth cells an attribution is positively faithful and anti-aligned with the true motif; ranking attributors by field-standard fidelity recovers the ground-truth-best method in distribution and fails under scaffold shift; and the ranking of both attributors and backbones is regime-specific rather than universal. MolSanity’s contribution is the audit that makes those failures visible — and, for the majority of real molecular cells, the explicit finding that correctness is not measurable at all with the labels the field currently has. The grid is still filling in; the pipeline, the figures and every number here regenerate from the committed artifacts as it does.

■ Data and code availability

All code, configurations, trained checkpoints, per-cell results and the scripts that generate every figure and table in this paper are available at <https://github.com/Kar488/molsanity>. The audit matrix is RESULTS.md, the head-to-head comparison BENCHMARK.md, and the faithfulness-versus-truth experiment BENCHMARK_GT.md / BENCHMARK_GT.json. This manuscript is built by paper/figs/make_figures.py and paper/figs/make_tables.py, which read those files directly; re-running them after further cells complete updates every number, figure and table in place. Dataset licences and provenance are recorded in the repository’s dataset manifest.

Table 5: Paired attributor comparisons on shared molecules (Wilcoxon signed-rank on per-molecule occlusion faithfulness). The powered in-distribution cell separates the attributor families decisively; the small shift cell mostly cannot, which is itself a reason not to rank attributors on faithfulness alone. p values are unadjusted for multiplicity and are printed at the precision of the committed artifact, in which 0.000 denotes $p < 0.0005$.

A vs. B	n	median Δ	p
<i>MUTAG, GINE, scaffold split</i>			
GNNExpl. vs. GuidedBP	20	-0.107	0.512
GNNExpl. vs. Input×Grad	20	-0.036	0.198
GNNExpl. vs. IG	20	-0.018	0.152
GNNExpl. vs. PGExpl.	20	0.200	0.020
GNNExpl. vs. Saliency	20	-0.036	0.233
GuidedBP vs. Input×Grad	20	0.000	0.196
GuidedBP vs. IG	20	0.000	0.211
GuidedBP vs. PGExpl.	20	0.268	0.123
GuidedBP vs. Saliency	20	0.000	0.559
Input×Grad vs. IG	20	0.000	1.000
Input×Grad vs. PGExpl.	20	0.468	0.052
Input×Grad vs. Saliency	20	0.000	0.102
IG vs. PGExpl.	20	0.468	0.047
IG vs. Saliency	20	0.000	0.206
PGExpl. vs. Saliency	20	-0.468	0.063
<i>SynthMotifsXL, GINE, random split</i>			
GNNExpl. vs. GuidedBP	120	-0.534	0.000
GNNExpl. vs. Input×Grad	120	-0.370	0.000
GNNExpl. vs. IG	120	-0.420	0.000
GNNExpl. vs. PGExpl.	120	0.278	0.000
GNNExpl. vs. Saliency	120	-0.398	0.000
GuidedBP vs. Input×Grad	120	0.126	0.000
GuidedBP vs. IG	120	0.101	0.000
GuidedBP vs. PGExpl.	120	0.807	0.000
GuidedBP vs. Saliency	120	0.093	0.000
Input×Grad vs. IG	120	-0.022	0.205
Input×Grad vs. PGExpl.	120	0.656	0.000
Input×Grad vs. Saliency	120	-0.025	0.000
IG vs. PGExpl.	120	0.684	0.000
IG vs. Saliency	120	-0.015	0.140
PGExpl. vs. Saliency	120	-0.689	0.000

Table 6: Molecular regression cells. Occlusion faithfulness is computed in output space (predicted-value shift) rather than probability space. In 16 of the 17 committed regression cells the attribution–occlusion agreement is *negative* — the atoms an attributor ranks highest are not the atoms whose removal moves the prediction most — the sole exception being Lipophilicity under GAT. Bold marks the highest occlusion ρ per dataset (emphasis only). PGExplainer is classification-only and therefore absent by construction.

dataset	backbone	attributor	split	n	RMSE	MAE	R^2	motif top-1	occ. ρ	occ. top-1	sparsity	stab.
ESOL	GAT	IG	scaffold	20	1.092	0.880	0.694	0.799	−0.481	0.150	0.730	0.922
ESOL	GCN	IG	scaffold	20	1.142	0.868	0.665	0.821	−0.820	0.100	0.720	0.967
ESOL	GINE	GNNExpl.	scaffold	20	1.132	0.918	0.671	0.828	−0.770	0.100	0.772	0.968
ESOL	GINE	GuidedBP	scaffold	20	1.132	0.918	0.671	0.659	−0.741	0.150	0.789	0.882
ESOL	GINE	Input×Grad	scaffold	20	1.132	0.918	0.671	0.881	−0.793	0.100	0.722	0.801
ESOL	GINE	IG	scaffold	20	1.132	0.918	0.671	0.834	−0.798	0.100	0.731	0.804
ESOL	GINE	Saliency	scaffold	20	1.132	0.918	0.671	0.878	−0.798	0.100	0.727	0.829
FreeSolv	GCN	IG	scaffold	20	2.567	2.048	0.538	0.782	−0.434	0.400	0.745	0.857
FreeSolv	GINE	GNNExpl.	scaffold	20	2.347	1.733	0.614	0.766	−0.600	0.350	0.755	1.000
FreeSolv	GINE	Input×Grad	scaffold	20	2.347	1.733	0.614	0.805	−0.370	0.450	0.714	0.754
FreeSolv	GINE	IG	scaffold	20	2.347	1.733	0.614	0.782	−0.582	0.400	0.738	0.943
FreeSolv	GINE	Saliency	scaffold	20	2.347	1.733	0.614	0.789	−0.425	0.450	0.736	0.843
Lipophilicity	GAT	IG	scaffold	20	0.940	0.746	0.391	0.762	0.458	0.900	0.785	0.812
Lipophilicity	GINE	GNNExpl.	scaffold	20	0.932	0.740	0.402	0.845	−0.499	0.000	0.788	0.867
Lipophilicity	GINE	Input×Grad	scaffold	20	0.932	0.740	0.402	0.699	−0.439	0.000	0.787	0.465
Lipophilicity	GINE	IG	scaffold	20	0.932	0.740	0.402	0.758	−0.149	0.000	0.785	0.889
Lipophilicity	GINE	Saliency	scaffold	20	0.932	0.740	0.402	0.705	−0.469	0.000	0.788	0.536

Table 7: Molecular classification cells (no node-level ground truth available), 1 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets — no per-atom labels are published — so only model-side reliability can be measured, which is precisely the gap the Tier-1 cells are needed to close. *charact.* is the GraphFrameEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules. Bold marks the best occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid−	stab.	charact.	unfaith.
BACE	GCN	IG	scaffold	30	0.133	0.224	0.398	0.776	0.766	0.900	0.597	0.597	0.873	0.370	0.619
BACE	GINE	GNNExpl.	scaffold	30	0.433	0.434	0.189	0.835	−0.128	0.167	0.064	0.214	0.932	0.223	0.036
BACE	GINE	Input×Grad	scaffold	30	0.433	0.434	0.189	0.841	−0.159	0.167	0.079	0.171	0.640	0.165	0.049
BACE	GINE	IG	scaffold	30	0.433	0.434	0.189	0.884	−0.252	0.167	0.074	0.202	0.386	0.200	0.047
BACE	GINE	Saliency	scaffold	30	0.433	0.434	0.189	0.873	−0.174	0.167	0.097	0.173	0.579	0.196	0.045
BBBP	GAT	IG	scaffold	30	0.967	0.888	0.024	0.810	−0.808	0.000	−0.069	−0.069	0.944	0.000	0.063
BBBP	GCN	IG	scaffold	30	0.833	0.888	0.016	0.781	−0.316	0.233	0.005	−0.096	0.883	0.085	0.049
BBBP	GINE	GNNExpl.	scaffold	30	0.967	0.805	0.059	0.823	−0.367	0.067	0.021	−0.077	0.929	0.102	0.053
BBBP	GINE	GuidedBP	scaffold	30	0.967	0.805	0.059	0.876	−0.530	0.000	−0.016	−0.037	0.890	0.061	0.084
BBBP	GINE	Input×Grad	scaffold	30	0.967	0.805	0.059	0.839	−0.620	0.000	−0.016	−0.047	0.897	0.063	0.110
BBBP	GINE	IG	scaffold	30	0.967	0.805	0.059	0.773	−0.430	0.000	0.006	−0.054	0.699	0.085	0.151
BBBP	GINE	PGExpl.	scaffold	30	0.967	0.805	0.059	0.422	−0.185	0.367	−0.055	−0.059	0.387	0.000	—
BBBP	GINE	Saliency	scaffold	30	0.967	0.805	0.059	0.826	−0.627	0.000	−0.029	−0.033	0.873	0.051	0.100
ClinTox	GINE	GNNExpl.	scaffold	30	0.567	0.820	0.207	0.483	0.927	0.700	0.375	0.356	0.963	0.246	0.394
ClinTox	GINE	Input×Grad	scaffold	30	0.567	0.820	0.207	0.793	0.139	0.567	0.356	0.356	0.583	0.246	0.388
ClinTox	GINE	IG	scaffold	30	0.567	0.820	0.207	0.724	0.085	0.633	0.356	0.356	0.859	0.246	0.386
ClinTox	GINE	Saliency	scaffold	30	0.567	0.820	0.207	0.790	0.145	0.567	0.356	0.356	0.570	0.246	0.382
DILI	GINE	GNNExpl.	scaffold	30	0.800	0.824	0.141	0.826	−0.044	0.433	0.080	0.154	0.949	0.109	0.110
DILI	GINE	Input×Grad	scaffold	30	0.800	0.824	0.141	0.858	−0.126	0.433	0.075	0.122	0.923	0.147	0.388
DILI	GINE	IG	scaffold	30	0.800	0.824	0.141	0.847	−0.101	0.433	0.073	0.128	0.897	0.151	0.386
DILI	GINE	Saliency	scaffold	30	0.800	0.824	0.141	0.851	−0.128	0.433	0.056	0.134	0.910	0.136	0.385

Table 8: Molecular classification cells (no node-level ground truth available), 2 of 2. Every committed classification cell on a real molecular dataset. The ground-truth localisation column does not exist for these datasets — no per-atom labels are published — so only model-side reliability can be measured, which is precisely the gap the Tier-1 cells are needed to close. *charact.* is the GraphFrameEx characterisation score and *unfaith.* the PyG/DIG unfaithfulness metric, computed on the same molecules. Bold marks the best occlusion ρ per dataset (emphasis only).

dataset	backbone	attributor	split	n	acc	AUC	ECE	motif top-1	occ. ρ	occ. top-1	Fid+	Fid−	stab.	charact.	unfaith.
SIDER	GCN	IG	scaffold	30	0.500	0.668	0.073	0.843	−0.512	0.333	0.014	−0.059	0.893	0.073	0.023
SIDER	GINE	GNNExpl.	scaffold	30	0.633	0.661	0.110	0.816	0.490	0.833	0.038	0.196	0.845	0.111	0.252
SIDER	GINE	Input×Grad	scaffold	30	0.633	0.661	0.110	0.878	0.576	0.867	0.088	0.196	0.724	0.176	0.499
SIDER	GINE	IG	scaffold	30	0.633	0.661	0.110	0.836	0.493	0.800	0.068	0.196	0.875	0.160	0.499
SIDER	GINE	Saliency	scaffold	30	0.633	0.661	0.110	0.887	0.571	0.867	0.093	0.201	0.753	0.180	0.498
Tox21	GINE	GNNExpl.	scaffold	30	1.000	0.827	0.054	0.719	0.010	0.367	−0.066	−0.061	1.000	0.002	0.171
Tox21	GINE	Input×Grad	scaffold	30	1.000	0.827	0.054	0.704	−0.409	0.167	−0.075	−0.060	0.809	0.001	0.211
Tox21	GINE	IG	scaffold	30	1.000	0.827	0.054	0.713	−0.339	0.200	−0.080	−0.062	0.664	0.001	0.211
Tox21	GINE	Saliency	scaffold	30	1.000	0.827	0.054	0.697	−0.391	0.167	−0.073	−0.063	0.760	0.002	0.208
hERG	GCN	IG	scaffold	30	0.900	0.802	0.098	0.665	0.711	0.867	0.589	0.590	0.910	0.368	0.641
hERG	GINE	GNNExpl.	scaffold	30	0.733	0.784	0.070	0.725	0.066	0.633	0.094	0.408	0.950	0.149	0.575
hERG	GINE	Input×Grad	scaffold	30	0.733	0.784	0.070	0.633	0.342	0.600	0.410	0.408	0.671	0.310	0.582
hERG	GINE	IG	scaffold	30	0.733	0.784	0.070	0.673	0.431	0.700	0.411	0.408	0.956	0.311	0.582
hERG	GINE	Saliency	scaffold	30	0.733	0.784	0.070	0.626	0.343	0.533	0.409	0.408	0.656	0.311	0.583

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